

Cluster-Scoped Routing for Fake-News Detection

Approach 2 — architecture, method, and results. A detector that discovers what *kind* of article it is reading with no domain label, routes it to a specialist trained on that kind, and shows similar labelled articles as evidence.

Primary corpus · FineFake (16,909)

Transfer test · ISOT (44,898)

Primary metric · macro-F1

Clusters · $k = 10$ (silhouette)

THE RESULT IN ONE GLANCE

0.773

Baseline macro-F1
(one global model
— the floor)

0.749

Approach 2 macro-F1
(clustered +
routed, no labels)

0.772

Ceiling macro-F1
(oracle routing by
true topic)

0.42

Approach 2 on
ISOT
(unseen corpus —
collapses)

Read plainly: on FineFake's own test set, label-free cluster routing lands **just below** the single global model and essentially level with the oracle ceiling — routing neither helps nor hurts much when the test data looks like training data. On a **totally unseen** corpus (ISOT) both systems fall off a cliff, and Approach 2 degrades slightly more gracefully than the baseline. The sections below build up to exactly why.

The pipeline, end to end

Approach 2 has two phases that share the same geometry. In the **training phase** the corpus is turned into vectors, grouped into clusters, and each cluster gets its own classifier. In the **inference phase** an unknown article travels the same vectoriser, is routed to the nearest cluster centroid, and is judged by that cluster's specialist — the "*closest cluster decides*" rule.

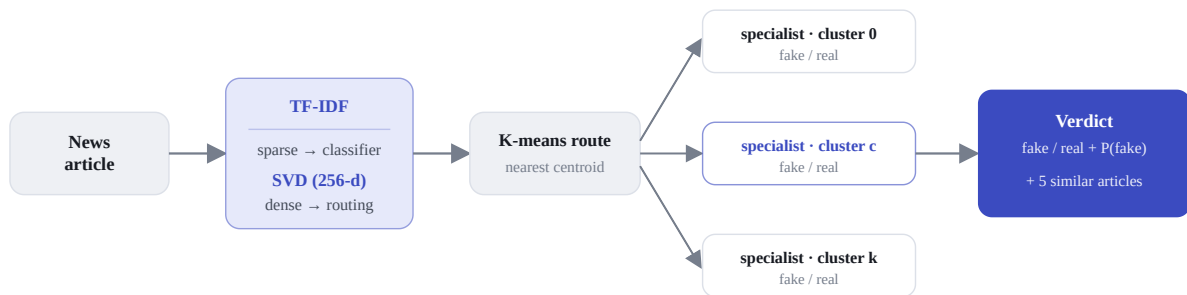


Figure 1. One article → two vector views → routed to its nearest cluster → that cluster's specialist returns a verdict, a confidence, and the five most similar labelled neighbours from the same cluster as evidence.

Building the clusters and their specialists

2.1 • Text → two vector representations

Every article is turned into numbers twice, because the two jobs want different things. A **sparse TF-IDF** vector (word 1–2 grams, 100,000 features) feeds the classifiers, so each learned weight is a literal word and the model stays readable. A **dense 256-dimension SVD** vector, L2-normalised, feeds the clustering and routing — compact, and its distances behave like cosine similarity, which also blunts FineFake's length confound (short articles skew fake, long ones skew real). Both are fit on **training rows only** to avoid leaking test vocabulary.

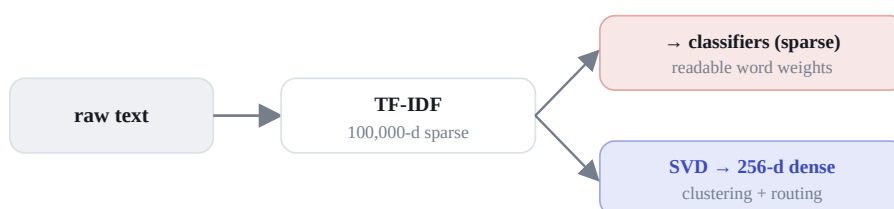


Figure 2. The same text yields a sparse view for the readable classifiers and a dense view for the geometry (clustering, routing, neighbour retrieval).

2.2 • Cluster with K-means, pick k by silhouette

K-means groups the dense training vectors. The number of clusters is not assumed — it is swept from 2 to 12 and the value with the best silhouette score is kept. On FineFake the curve rises to a clear peak at **k = 10** (note this is not forced to match FineFake's 7 true topics — the grouping is purely geometric).

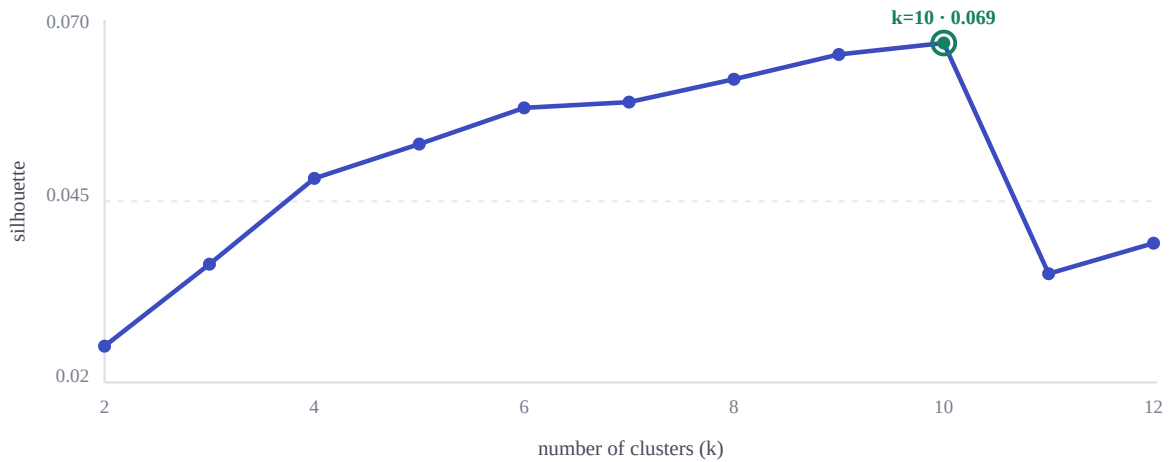


Figure 3. Silhouette score across candidate k on the real FineFake training set. The best value, $k = 10$, is selected automatically.

2.3 · Are the clusters clean topics? (purity = 0.393)

The clusters are compared against the true topics *only as a diagnostic* — they never use the labels. The result is telling: cluster purity is just **0.393**, and the majority topic of eight of ten clusters is "Politics" simply because Politics is a third of the corpus. Only cluster 0 (Entertainment) is a clean, distinctive group. This low purity is the single most important fact for understanding the results.

Table 1 — Discovered clusters: size and dominant true topic (diagnostic only)

CLUSTER	TRAIN ROWS	TEST ROWS	MAJORITY TOPIC	NOTE
0	1,150	258	Entertainment	clean, distinctive
1	304	51	Politics	small, mixed
2	611	143	Politics	mixed
3	339	78	Society	small, mixed
4	2,727	599	Politics	large, generic
5	377	103	Politics	mixed
6	3,526	659	Politics	largest, generic
7	443	94	Politics	mixed
8	2,089	486	Politics	large, mixed
9	270	66	Society	small, mixed
Total	11,836	2,537	purity 0.393	—

2.4 • One specialist classifier per cluster

Each cluster trains its own logistic-regression classifier on the sparse TF-IDF rows that landed in it, so it can learn the writing patterns of that group. A cluster too small (< 100 rows) or single-class would fall back to a global classifier — on this run all ten clusters were trainable. The idea is the report's picture below: each group gets its own decision rule, so no single line has to fit everything.

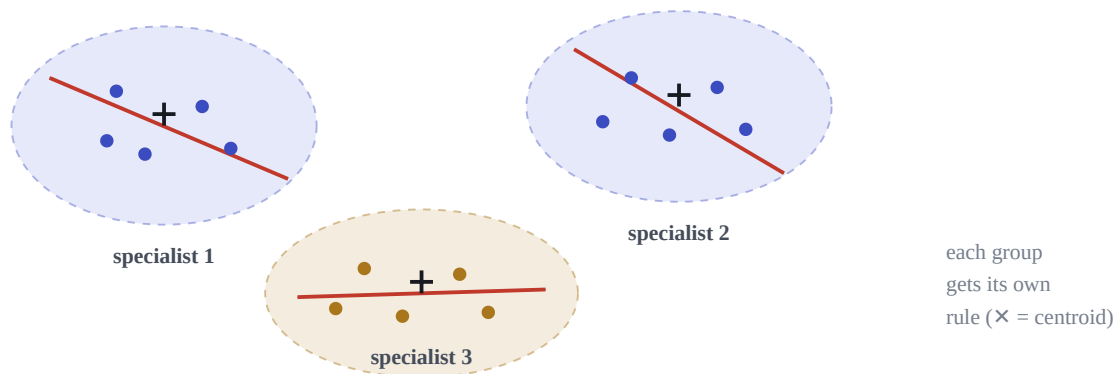


Figure 4. Clusters formed from geometry, each with its own specialist decision boundary (red). The X marks each cluster centroid — the anchor the router measures distance to.

03 – Inference phase

Routing an unknown article

A new article arrives with no label. It is vectorised through the *same* fitted TF-IDF and SVD, and `kmeans.predict()` assigns it to the nearest cluster centroid. That cluster's specialist produces the verdict and a confidence $P(\text{fake})$. Because routing and evidence use the same dense vectors, the system can also return the five most similar labelled articles from that cluster — the explanation is produced by the same geometry that produced the decision.

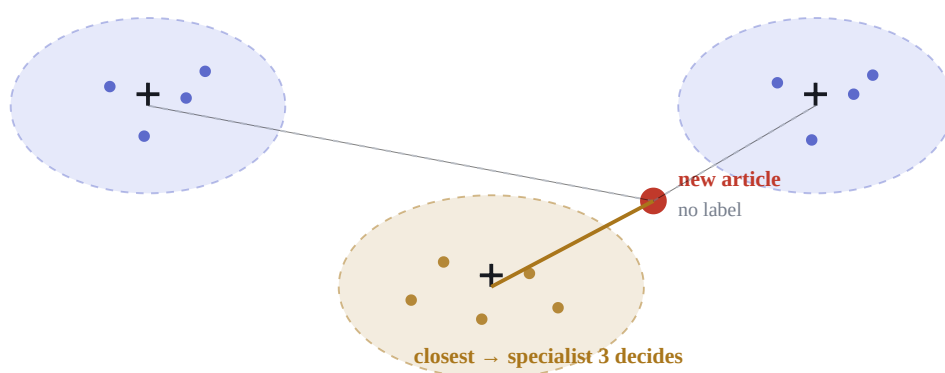


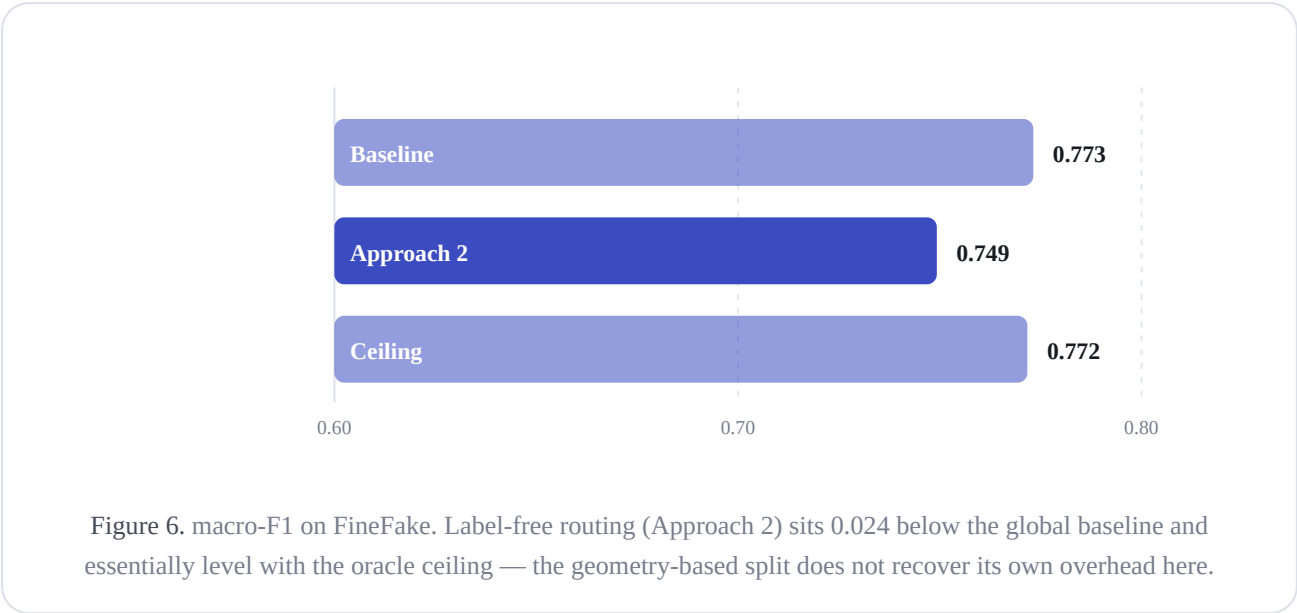
Figure 5. The unlabelled article is assigned to the cluster whose centroid is nearest in the dense SVD space; that cluster's specialist returns the verdict, and its neighbours become the evidence.

On FineFake's own test set

All three systems share one vectoriser fitted on the training rows, so the comparison isolates the routing mechanism. macro-F1 is primary; the corpus is ~1.25 : 1 fake : real, so accuracy alone flatters a lazy guesser.

Table 2 — FineFake test set (2,537 articles). Positive class = fake.

SYSTEM	MACRO-F1	ACCURACY	PRECISION	RECALL	
Baseline — one global model floor	0.7732	0.7781	0.783	0.831	0
Approach 2 — clustered + routed	0.7494	0.7572	0.753	0.839	0
Ceiling — oracle topic routing	0.7715	0.7770	0.778	0.838	0



4.1 · Per-cluster breakdown

The pooled 0.749 hides wide variation. Cluster 0 (clean Entertainment) is perfect; the small, mixed clusters 1 and 3 fall below 0.50 and drag the mean down. This is exactly what low purity predicts: a specialist trained on a muddled group is a muddled specialist.

Table 3 — Approach 2, per-cluster (routed test articles)

CLUSTER	N TEST	MACRO-F1	ACCURACY	ROC-AUC	MAJORITY TOPIC
0	258	1.0000	1.000	1.000	Entertainment
1	51	0.4784	0.647	0.659	Politics
2	143	0.6454	0.685	0.723	Politics
3	78	0.3983	0.590	0.448	Society
4	599	0.5990	0.790	0.656	Politics
5	103	0.6400	0.709	0.651	Politics
6	659	0.6765	0.744	0.778	Politics
7	94	0.5428	0.596	0.581	Politics
8	486	0.6963	0.704	0.780	Politics
9	66	0.7326	0.788	0.644	Society
Pooled	2,537	0.7494	0.757	0.838	—

WHY POOLED > MOST CLUSTERS

The pooled macro-F1 (0.749) is higher than most per-cluster scores because small clusters have skewed class balance that hurts their *local* F1, while the pooled figure is dominated by the large clusters and the perfect cluster 0. Both views are true; report them together.

On ISOT — a corpus never seen in training

The FineFake-trained baseline and Approach-2 router were run, unchanged, over all 44,898 ISOT articles. ISOT's labels are source-level (real = Reuters, fake = flagged sites) and its articles are ~20× longer (median 375 vs 20 words), so this is a hard distribution shift.

Table 4 — FineFake → ISOT transfer (nothing retrained on ISOT)

SYSTEM	IN-CORPUS F1	ISOT F1	Δ	ISOT ACC	FAKE RECALL
Baseline — global	0.7732	0.3955	-0.378	0.507	0.074
Approach 2 — routed	0.7494	0.4185	-0.331	0.504	0.116

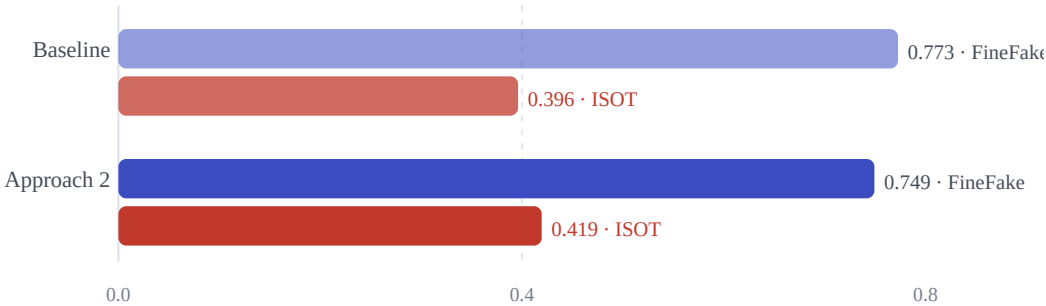


Figure 7. Both systems drop from ~0.75 to ~0.40 on the unseen corpus. Approach 2 loses slightly less (-0.331 vs -0.378) and catches more fakes (recall 0.116 vs 0.074).

5.1 · The router collapses under distribution shift

A revealing diagnostic: **90.6%** of ISOT articles (40,668 of 44,898) route to a single FineFake cluster (cluster 4). ISOT's long, uniform Reuters-style prose all looks alike to the FineFake-fitted geometry, so "route to the cluster you resemble" stops discriminating — routing effectively degenerates to one specialist. This is itself a robustness finding, not just a low score.

Table 5 — Where ISOT articles route (FineFake clusters)

CLUSTER	ISOT ARTICLES	SHARE
4	40,668	90.6%
8	1,864	4.2%
7	1,233	2.7%
6	735	1.6%
9	249	0.6%
2	105	0.2%
1	42	0.1%
5	2	<0.1%
0, 3	0	0.0%

What the numbers mean

- **In-corpus, routing doesn't beat a single model (−0.024).** With cluster purity at 0.393, the discovered groups are not clean topics, so the specialists are trained on mixed data and the cost of splitting isn't recovered. This echoes Approach A: even oracle topic routing only matched the baseline here.
- **Approach 2 is marginally more robust on transfer (+0.023 macro-F1, and fake recall 0.116 vs 0.074).** Routing didn't help when test looked like train, but it helped a little when the data was foreign — a genuine, if small, point in its favour.
- **Neither model generalises to ISOT** (both ≈ 0.40 , ROC-AUC ≈ 0.58). This is the project's central thesis made concrete: a detector trained on one corpus does not transfer to another.
- **The router degenerates under shift** — 90.6% of ISOT to one cluster. Nearest-centroid routing assumes new data falls near the training geometry; when it doesn't, routing carries no information.

CAVEATS FOR THE WRITE-UP

ISOT's ground truth is source-level (weaker than FineFake's), and its documents are $\sim 20\times$ longer — so part of the collapse is style/length mismatch, not only topic shift. And the whole project is fabrication detection, not fact verification: it reads how an article is written, never whether the claim is true.

Together these motivate **Approach 3**: replace the fixed geometric partition with a *learned* one (a gated mixture of experts) where the routing is trained from the classification loss rather than raw distance — so the split is judged by whether it helps prediction, not by how the vectors happen to cluster.

Exact settings

Table 6 — Hyperparameters and data	
COMPONENT	SETTINGS
Split	70 / 15 / 15, stratified on (label, topic), seed 42 — the committed split
TF-IDF (classifier)	word 1-2 grams, min_df 3, max 100,000 features, sublinear
SVD (routing)	256 components, L2-normalised (cosine)
Clustering	K-means, k swept 2-12, k = 10 by silhouette, n_init 10, max_iter 1000
Specialist	logistic regression, max_iter 2000, class_weight balanced
Fallback	clusters < 100 rows → global classifier (none trained)
FineFake	16,909 rows · 9,407 fake / 7,502 real · median 200 words
ISOT (transfer)	44,898 rows · 23,481 fake / 21,417 real · median 37 words

Reproduce in the repo: run notebooks/02_approach_b.ipynb (loads FineFake, clusters, routes, writes results/approach_b.csv and saves the fitted models), then python scripts/eval_isot_transfer.py for the ISOT transfer table. No existing project files are modified.

Approach 2 · Cluster-Scoped Routing for Fake-News Detection · Advanced Data Mining, Phase A, Group E. All figures computed on the real FineFake and ISOT corpora. Fabrication-detection system; output is a triage probability, not a verdict on truth.